

15. (2.25 points)

An actuary is retrospectively testing the general applicability of the Mack stochastic reserving model for Products Liability using incurred loss triangles from 20 insurance companies.

The following procedure was performed:

- The Mack model was used to fit each insurer's incurred loss triangle as of 10 years before the latest available annual data.
- A lognormal distribution was derived for each triangle by matching the mean and standard deviation from the Mack model results.
- Percentiles for each insurer's actual 10-year loss emergence were calculated assuming the lognormal distributions described above.

Given the following percentiles for each company's actual loss development within the distribution derived with the Mack model, in ascending order:

12	45	53	67
29	49	58	69
37	50	61	71
40	51	62	72
44	52	66	84

a. (1 point)

Determine whether the model is validated by applying the Kolmogorov-Smirnov test at the 5% confidence level, where the critical value is 30.4.

b. (0.5 point)

Identify two reasons why loss reserve models often do not accurately predict the distribution of outcomes.

c. (0.75 point)

Identify one shortcoming of the Mack model. Propose an alternative model and briefly explain a feature of the alternative model that addresses this shortcoming.

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QUESTION 15

TOTAL POINT VALUE: 2.25

LEARNING OBJECTIVE: A9: Test assumptions underlying reserve models.

SAMPLE ANSWERS

Part a: 1.0 point

Sample Answer 1

Actual percentiles (p_i)	Expected Percenties (f_i)	$ p_i - f_i $
12	$100 \times (1/n) = 100 \times (1/20) = 5$	7
29	$100 \times (2/n) = 10$	19
37	15	22
40	20	20
44	25	19
45	30	15
49	35	14
50	40	10
51	45	6
52	50	2
53	55	2
58	60	2
61	65	4
62	70	8
66	75	9
67	80	13
69	85	16
71	90	19
72	95	23
84	100	16

- $\text{Max}|p_i - f_i| = 23 < 30.4$ therefore the model is validated

Sample Answer 2

$p =$ percentile	$e = [x / (n + 1)]$ $= [x / 21]$	$D = p - e $
0.12	0.0476	0.0724
0.29	0.0952	0.1948
0.37	0.1429	0.2271
0.40	0.1905	0.2095
0.44	0.2381	0.2019
0.45	0.2857	0.1643
0.49	0.3333	0.1567
0.50	0.3810	0.1190

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	0.51	0.4286	0.0814	
	0.52	0.4762	0.0438	
	0.53	0.5238	0.0062	
	0.58	0.5714	0.0086	
	0.61	0.6190	0.0090	
	0.62	0.6667	0.0467	
	0.66	0.7143	0.0543	
	0.67	0.7619	0.0919	
	0.69	0.8095	0.1195	
	0.71	0.8571	0.1471	
	0.72	0.9048	0.1848	
	0.84	0.9524	0.1124	
None of the Ds are larger than 0.304 so the model is validated at 5% confidence level.				
Part b: 0.5 point				
<i>Sample Answers (any two of which would earn full credit)</i> - There is another model that would work better. - Insurance loss environment has experienced changes that are not yet observable in the data. - Insurance process is too dynamic to be captured by a single model. OR Black swan type events can distort the modeled process. The insurance process is just too complex. - Underlying data used to calibrate the model is missing crucial info necessary to make a reliable prediction. - Because we only use a small sample of universe data we are likely to miss-estimate parameters.				
Part c: 0.75 point				
<i>Sample Answer 1</i> - The Mack model uses the reported to date cumulative loss to estimate the ultimate loss level for AY. The reported to date is fixed and thus the row level acts as a fixed parameter. - By allowing the row level to vary, we can add more volatility to the Mack model (which produced light tails on incurred data). The Levelled Chain Ladder model does this.				
<i>Sample Answer 2</i> - Mack assumes that AYs are independent. - If we remove this assumption and allow AY to be correlated and vary the level of the prior loss we can produce the CCL (Correlated Chain Ladder) model which passes the K-S test due to adding increased variation in the loss projection.				
<i>Sample Answer 3</i> - One shortcoming of the Mack model is it only provides mean and variance, not a full distribution. - An alternative is the ODP Bootstrap as its output provides a full distribution in the form of the simulated results.				

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Sample Answer 4

- A shortcoming of Mack model is it doesn't capture the speeding settlement rate in today's environment.
- We can use the Changing Settlement Rate (CSR) model instead to allow incorporation of a variable to represent settlement rate and let it increase over time.

Sample Answer 5

- One shortcoming of the Mack model is that it doesn't allow for incorporating expert opinion and calculate a prediction error around an incorporated expert opinion.
- An alternative is a Bayesian credibility model which allows for incorporation of expert opinions and valid prediction errors estimates.

EXAMINER'S REPORT

- Candidates were expected to know the main ideas presented in the Meyers paper.
- The candidates in general scored well on part a, relatively well on part c, and not very well on part b.
- The biggest problem candidates had on b. and c. was to distinguish between a shortcoming itself and the result of such.
- Also, Meyers' language in describing his findings was confusing to the candidates. Meyers commonly refers to low standard deviation as "light-tail" feature of the distribution (and Mack doesn't give the distribution, only its mean and variance). Same with "biased high" feature of the distribution describing the mean of model distribution being higher than the actual mean. These two descriptions appeared in almost two-thirds of the answers (to both b. and c.) and very often they were not explained, and very often given as shortcomings to either reserving models in general or to Mack's model in general.

Part a

Candidates were expected to know how to perform the K-S test over a uniform distribution, with the critical value given.

- Candidates in general scored well on this
- Candidates were allowed to use n or $n+1$ in the denominator for uniformly distributed percentiles (f_i). We accepted both answers, following Meyers. When using $n+1$ as the denominator, the most common error was not to use the absolute value in the test (the biggest absolute value had a positive difference).

Part b

Candidates were expected to know common challenges to reserving models.

- Common errors were providing answers that were too general or too similar. For example, some candidates responded with "could be biased high" and "could be biased low", which are not distinct challenges to reserving models.
- Another common error (and related to the one above) was confusing the model features or challenges to the model with the results of the model.

Part c

Candidates were expected to know the shortcomings to Mack's model and how they could be overcome by other models (either proposed by Meyers or in other papers)

- The most common error was to confuse the result of the shortcoming ("light-tail", "bias high") with the shortcoming itself (such as ignoring CY effects). Also, not understanding that the resulting "light-tails" or "high bias" were present because of the features of the

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data set on which the model was tested. One cannot make generalizations about the model when testing it on one set of data only, even if the data is fully credible, but generalizations were made very often (including not being specific about in which types of triangles, incurred or paid, the effect was observed).

- While candidates could generally propose a different model, they often struggled to describe why that model represented an improvement.